THE COLLISION SPECTRUM AND THE L-FUNCTION LANDSCAPE

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Abstract. The collision transform decomposes the centered collision deviation S° over odd Dirichlet characters modulo $m = b^{\ell+1}$. This paper proves that at lag 1 (so $m = b^2$), for prime b and primitive odd χ:

$$\hat{S}^\circ(\chi) = -\frac{B_{1,\overline{\chi}} \cdot \overline{S_G(\chi)}}{\phi(m)},$$

where $B_{1,\overline{\chi}}$ is the generalized Bernoulli number and $S_G(\chi)$ is the diagonal character sum. In magnitude: $|\hat{S}^\circ| = |B_1| \cdot |S_G|/\phi(m)$.

Combined with the Bernoulli–L-value formula

$$|B_1| = \frac{\sqrt{m}}{\pi}\,|L(1,\chi)|$$

and the S_G factorization theorem (proved in a companion paper [5]), this gives

$$|\hat{S}^\circ(\chi)| = \frac{\sqrt{m}}{\pi\,\phi(m)}\,|L(1,\chi)| \cdot |S_G(\chi)|.$$

The collision invariant's Fourier spectrum encodes L-function special values. Computation shows $|S_G|$ itself correlates with $|L(1)|$, giving $|\hat{S}^\circ| \sim |L(1)|^\alpha$ with $\alpha = 2.00$ at base 5 (exact) and $\alpha \approx 1.80$ across the tested prime bases.

1. Introduction

The collision transform [4] decomposes the centered collision deviation S° over Dirichlet characters modulo $m = b^{\ell+1}$. The antisymmetry theorem restricts the decomposition to odd characters. The preceding papers recorded computationally that the collision coefficients $|\hat{S}^\circ(\chi)|$ correlate with $|L(s,\chi)|$ at Pearson $r > 0.87$ across all prime bases tested [6, 7].

This paper proves the structural identity underlying the correlation. The proof uses three ingredients: the slice formula from [3], the classical Bernoulli identity for character sums over fractional parts, and the vanishing of coset sums for primitive characters.

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2. The Decomposition Theorem

Theorem 1 (Decomposition)**.** *Let b be prime, $m = b^2$, and χ a primitive odd character modulo m. Then*

$$\hat{S}^\circ(\chi) = -\frac{B_{1,\overline{\chi}} \cdot \overline{S_G(\chi)}}{\phi(m)},$$

where $B_{1,\overline{\chi}} = (1/m)\sum_a a\,\overline{\chi}(a)$ is the generalized first Bernoulli number and $G = \{r(b{+}1) : 0 \le r \le b{-}1\}$ is the diagonal set, and $S_G(\chi) = \sum_{n\in G}[\overline{\chi}(n{+}1) - \overline{\chi}(n)]$ is the diagonal character sum.

Proof. Write $\phi = \phi(m)$. The Fourier coefficient is

$$\phi\,\hat{S}^\circ(\chi) = \sum_a S^\circ(a)\,\overline{\chi}(a) = \sum_a \big[S(a) - \overline{S}_{R(a)}\big]\,\overline{\chi}(a).$$

Step 1: the centering term vanishes. The class mean $\overline{S}_R$ is constant on each spectral class $R = (a{-}1) \bmod b$. The sum over each class is $\sum_{a\equiv k \bmod b}\overline{\chi}(a) = \overline{\chi}(k)\sum_{j=0}^{b-1}\overline{\chi}(1+jb)$. For primitive χ modulo b^2 (conductor b^2, not b), the restriction of χ to the subgroup $\{1+jb : j = 0,\dots,b{-}1\}$ is non-trivial. By character orthogonality over this subgroup, the sum vanishes. Therefore $\sum_a \overline{S}_{R(a)}\,\overline{\chi}(a) = 0$.

Step 2: expand $S(a)$ via the slice formula. By [3], $S(a) = -1 - \lfloor a/b\rfloor + \sum_{n\in G} d_n(a)$ where $d_n(a) = \lfloor (n{+}1)a/m\rfloor - \lfloor na/m\rfloor$.

The constant term: $\sum_a(-1)\,\overline{\chi}(a) = 0$ (non-trivial χ).

The floor term: $\lfloor a/b\rfloor = a/b - \{a/b\}$. Since $\{a/b\} = \{ba/m\}$ and the Bernoulli identity (Lemma 2 below) gives $\sum_a\{ba/m\}\,\overline{\chi}(a) = \chi(b)\,B_{1,\overline{\chi}} = 0$ (because $\gcd(b,m) \ne 1$, so $\chi(b) = 0$):

$$-\sum_a \lfloor a/b\rfloor\,\overline{\chi}(a) = -(1/b)\sum_a a\,\overline{\chi}(a) + 0 = -b\,B_{1,\overline{\chi}}.$$

Step 3: the diagonal terms. The endpoint slices are handled separately. For $n = 0$, one has $d_0(a) = 0$, so the contribution vanishes. For $n = m{-}1$, one has $d_{m-1}(a) = 1$ for every unit a, so the sum against the non-trivial character $\overline{\chi}(a)$ also vanishes.

For the interior slices, Lemma 2 gives $\sum_a\lfloor na/m\rfloor\,\overline{\chi}(a) = [n - \chi(n)]\,B_{1,\overline{\chi}}$ for $\gcd(n,m) = 1$. Therefore

$$\begin{aligned}\sum_a d_n(a)\,\overline{\chi}(a) &= [(n{+}1) - \chi(n{+}1)]\,B_1 - [n - \chi(n)]\,B_1\\ &= [1 + \chi(n) - \chi(n{+}1)]\,B_1.\end{aligned}$$

Summing over G ($|G| = b$ slices):

$$\sum_{n\in G}\sum_{a} d_n(a)\,\overline{\chi}(a) = B_1\left[b - \sum_{n\in G}(\chi(n{+}1) - \chi(n))\right] = B_1\big[b - \overline{S_G(\chi)}\big].$$

Step 4: combine.

$$\phi\,\hat{S}^\circ(\chi) = 0 + (-b\,B_1) + B_1\big[b - \overline{S_G(\chi)}\big] - 0 = -B_1 \cdot \overline{S_G(\chi)}.$$

Dividing by ϕ gives the result. □

Lemma 2 (Bernoulli identity)**.**
For primitive χ modulo m and $\gcd(n, m) = 1$*:*

$$\sum_{\substack{a=1\\ \gcd(a,m)=1}}^{m-1} \overline{\chi}(a)\left\{\frac{na}{m}\right\} = \chi(n)\,B_{1,\overline{\chi}}.$$

Proof. The fractional part $\{na/m\}$ is periodic in a modulo m. Since $\gcd(n, m) = 1$, the map $a \mapsto na$ permutes the units, so

$$\begin{aligned}\sum_{a}\overline{\chi}(a)\{na/m\} &= \sum_{a}\overline{\chi}(n^{-1}a)\{a/m\}\\ &= \chi(n)\sum_{a}\overline{\chi}(a)\{a/m\} = \chi(n)\,B_{1,\overline{\chi}},\end{aligned}$$

where the last step uses $B_{1,\overline{\chi}} = (1/m)\sum a\,\overline{\chi}(a) = \sum\overline{\chi}(a)\,\{a/m\}$ (since $\{a/m\} = a/m$ for $1 \le a \le m{-}1$). □

3. The L-Function Encoding

Corollary 3. *For prime b and primitive odd χ modulo b^2:*

$$|\hat{S}^\circ(\chi)| = \frac{\sqrt{m}}{\pi\,\phi(m)}\,|L(1,\chi)| \cdot |S_G(\chi)|.$$

Proof. By the Bernoulli–L-value formula for odd primitive characters [1], $|B_{1,\overline{\chi}}| = \sqrt{m}\,|L(1,\chi)|/\pi$. Substituting into $|\hat{S}^\circ| = |B_1| \cdot |S_G|/\phi$ gives the result. □

The collision coefficient factors into:

- $|L(1,\chi)|$: the L-function special value, measuring distance from zeros;
- $|S_G(\chi)|$: the diagonal character sum, which by the S_G factorization theorem [5] equals $2\,|\sum_{k=1}^{b-1}\overline{\chi}(k)|$.

4. The Double Encoding

The S_G factorization gives $|S_G| = 2|\sum_{k=1}^{b-1} \overline{\chi}(k)|$. Computation shows this partial sum correlates with $|L(1,\chi)|$.

b	n	$r(\lvert P\rvert, \lvert L\rvert)$	$r \cdot \log b$
5	8	1.00	1.61
7	18	0.88	1.71
13	72	0.83	2.13
19	162	0.86	2.52
31	450	0.84	2.90
43	882	0.85	3.18
61	1800	0.84	3.46
71	2450	0.84	3.57

Table 1. Correlation between the partial sum $|P| = |\sum_{k=1}^{b-1} \overline{\chi}(k)|$ and $|L(1,\chi)|$ across primitive odd characters. The correlation is consistent with decay on the scale of $c/\log b$.

At base 5, the encoding is exact: $|P| = (\sqrt{5}/2)\,|B_{1,\overline{\chi}}|$ for every primitive character, giving $|\hat{S}^\circ| \propto |L(1)|^2$. At larger bases, the correlation decays, and the effective exponent in $|\hat{S}^\circ| \sim |L(1)|^\alpha$ decreases from 2.00 toward 1.

5. The Apostol Decomposition

The partial sum $P = \sum_{k=1}^{b-1} \overline{\chi}(k)$ has an exact decomposition via the Apostol–Berndt identity for short character sums [1]. Normalizing by the Gauss sum:

$$\widetilde{P} = \frac{\pi}{\tau(\chi)}\, P = L(1,\overline{\chi}) + \Delta(\chi),$$

where the *packet*

$$\Delta(\chi) = \frac{i}{\varphi(b)} \sum_{\substack{\xi \bmod b \\ \xi(-1)=1,\, \xi\neq 1}} \tau(\overline{\xi})\, L(1,\xi\overline{\chi})$$

is a sum of twisted L-values weighted by Gauss sums over the even characters modulo b.

Observation 4 (Packet magnitude)**.** The mean ratio $|\Delta|/|L(1)|$ is constant at approximately 1.08 across all prime bases tested ($b = 5$ to 43). The packet has the same magnitude as the main term.

Observation 5 (Phase uniformity)**.** The relative phase $\theta = \arg(\Delta) - \arg(L(1, \overline{\chi}))$ has $\langle \cos\theta \rangle = 0.000$ at every base. The packet's phase is uniformly distributed relative to $L(1)$.

Observation 6 (Variance decay)**.** The standard deviation of $|\Delta|/|L(1)|$ across primitive odd characters is consistent with decay on the scale of $c/\log b$, with $\mathrm{std} \cdot \log b \approx 1.2$ across all bases tested.

A natural explanation for the correlation decay is the observed packet behavior. The partial sum $\widetilde{P} = L(1) + \Delta$ combines two terms of equal magnitude and uniform relative phase. When the ratio $|\Delta|/|L|$ has high variance (small b), the variation in $|\widetilde{P}|$ tracks the variation in $|L|$, producing high correlation. As b grows, the variance of $|\Delta|/|L|$ decreases (the packet becomes more uniform relative to L), and the L-specific information in $|\widetilde{P}|$ is washed out.

This is consistent with the fact that Δ is a sum of $(\varphi(b)/2-1)$ twisted L-value terms. As the number of terms grows, the relative fluctuation of the sum decreases, concentrating $|\Delta|/|L|$ around its mean.

6. The Moment Theorem

The decomposition theorem translates the primitive odd sector of the collision transform into a mean-value statement about L-function special values. Write

$$F^{\circ}_{\mathrm{prim}}(s) = \sum_{\substack{\chi \bmod m \\ \chi \text{ primitive odd}}} \hat{S}^{\circ}(\chi)\, P(s, \chi), \qquad m = b^2.$$

Theorem 7 (Moment majorant)**.** *For prime b and primitive odd characters modulo $m = b^2$, one has*

$$\left|F^{\circ}_{\mathrm{prim}}(1)\right| \le \frac{1}{\phi(m)} \sum_{\substack{\chi \bmod m \\ \chi \text{ primitive odd}}} |B_{1,\overline{\chi}}| \cdot |S_G(\chi)| \cdot |P(1, \chi)|$$

where $P(1, \chi) = \sum_p \chi(p)/p$.

Proof. By the decomposition theorem,

$$F^{\circ}_{\mathrm{prim}}(1) = -\frac{1}{\phi} \sum_{\chi \text{ prim. odd}} B_{1,\overline{\chi}} \cdot \overline{S_G(\chi)} \cdot P(1, \chi).$$

The triangle inequality gives the stated majorant. □

By the Bernoulli–L-value formula, $|B_1| = \sqrt{m}\, |L(1, \chi)|/\pi$. By the S_G factorization, $|S_G| = 2\,|\sum_{k=1}^{b-1} \overline{\chi}(k)|$. And $P(1, \chi)$ relates to $\log L(1, \chi)$ via the Euler product. The majorant therefore places $|F^{\circ}_{\mathrm{prim}}(1)|$ beneath

an explicit weighted combination of $|L(1,\chi)| \cdot |S_G(\chi)| \cdot |\log L(1,\chi)|$ across the primitive odd characters modulo m.

Corollary 8. *At lag* 1*, the modulus* $m = b^2$ *has* $\phi(m)/2 = b(b-1)/2$ *odd characters. The moment majorant expresses* $|F^\circ_{\text{prim}}(1)|$ *as lying beneath an explicit weighted average over the primitive odd family.*

Remark 9. At base 5, lag 1: $|S_G| = (5\sqrt{5}/\pi)|L(1)|$ exactly, so the moment majorant reads

$$\left|F^\circ_{\text{prim}}(1)\right| \le C \sum |L(1)|^2 |\log L(1)|$$

for an explicit constant C: a weighted second moment majorizing the finite collision transform at the edge.

Theorem 10 (Conditional cross-moment)**.** *Fix* $\sigma_0 \in (1/2, 1)$ *and suppose every primitive odd* L*-function modulo* $m = b^2$ *satisfies* $L(s,\chi) \neq 0$ *for* $\mathrm{Re}(s) > \sigma_0$*. Then for every real* $s > \sigma_0$*:*

$$\left|F^\circ_{\text{prim}}(s)\right| \le \frac{1}{\phi(m)} \sum_{\substack{\chi \bmod m \\ \chi \text{ primitive odd}}} |B_{1,\overline{\chi}}| \cdot |S_G(\chi)| \cdot |P(s,\chi)|.$$

Proof. By the conditional penetration theorem [4], $F^\circ(s)$ converges for real $s > \sigma_0$. The decomposition theorem gives $F^\circ_{\text{prim}}(s) = -(1/\phi) \sum B_1 \cdot \overline{S_G} \cdot P(s,\chi)$. The triangle inequality gives the bound. □

Since $|B_1| = \sqrt{m}\,|L(1,\chi)|/\pi$ and $P(s,\chi) = \log L(s,\chi) - H(s,\chi)$, the cross-moment majorant places $|F^\circ_{\text{prim}}(s)|$ beneath a weighted sum of products

$$|L(1,\chi)| \cdot |S_G(\chi)| \cdot |\log L(s,\chi)|$$

across odd characters. This connects L-values at two points: $s = 1$ (the edge, through B_1) and $s = \sigma_0$ (in the strip, through P). As $\sigma_0 \to 1/2$, the bound constrains L-values approaching the critical line, weighted by L-values at the edge and filtered by the collision geometry through S_G.

7. The Zero Coupling

The decomposition theorem gives the exact structural identity behind part of what the preceding papers [6, 7] observed. The factor B_1 shows that characters with small $|L(1,\chi)|$ necessarily have small collision coefficients $|\hat{S}^\circ(\chi)|$. This does not by itself prove the anti-correlation, the spectral repulsion, or the double transversality, since those involve additional information about $S_G(\chi)$, partial prime sums, and cross-character geometry. It does, however, explain why the edge value $L(1,\chi)$ is a structural input to every coefficient.

The analytic continuation from [4] gives

$$\mathcal{F}^\circ(s) = -\frac{1}{\phi} \sum_{\chi \text{ odd}} B_{1,\overline{\chi}} \cdot \overline{S_G(\chi)} \cdot \big[\log L(s,\chi) - H(s,\chi)\big].$$

Near a zero ρ of $L(s,\chi_0)$, the term $\log L(s,\chi_0) \to -\infty$. The coefficient of this divergence is $B_{1,\overline{\chi}_0} \cdot \overline{S_G(\chi_0)}/\phi$, which involves $L(1,\chi_0)$ through B_1. The behavior of the collision transform near a zero at $s = \rho$ is therefore controlled by the L-value at $s = 1$.

Remark 11. The digit function couples different points of the critical strip: the singularity structure at depth s is weighted by the health at $s = 1$. A zero of $L(s,\chi)$ deep in the strip produces a singularity in $\mathcal{F}^\circ$ whose residue is proportional to $|L(1,\chi)|$. Characters that are healthy at the edge produce large residues at depth. Characters that are weak at the edge produce small residues. The collision geometry mediates this coupling through the Bernoulli number.

8. Remarks

The decomposition $\hat{S}^\circ = -B_1 \cdot \overline{S_G}/\phi$ is exact for every primitive odd character at every prime base. It connects the collision invariant to two classical objects: the Bernoulli number B_1 (encoding $|L(1)|$ via the Bernoulli–L-value formula) and the partial character sum $P = \sum_{k=1}^{b-1} \overline{\chi}(k)$ (encoding a second, decaying factor of $|L|$).

At base 5, the double encoding is exact: $|\hat{S}^\circ| \propto |L(1)|^2$. At larger bases, the second factor's encoding degrades logarithmically. The digit function's "opening act" (its values at the first $b-1$ units) carries information about the L-function landscape, but the fidelity of this information is consistent with decay on the scale of $1/\log b$ across the tested prime bases.

For imaginary quadratic fields $\mathbb{Q}(\sqrt{-d})$, the class number satisfies $h(-d) = (\sqrt{d}/\pi)\,|L(1,\chi_d)|$. The decomposition theorem therefore encodes class numbers: at base 5, $|\hat{S}^\circ| \propto h(-d)^2$. The digit function squares class numbers.

The open problem is to prove this decay rate. The decomposition theorem, the Abel summation formula, and the near-linear structure of $G(k)$ provide the algebraic framework. What remains is a bound on the correlation between partial character sums over $\{1,\ldots,b-1\}$ and L-function special values for primitive characters modulo b^2.

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